



8:45 - 9:00 Welcome

9:00 - 10:00 Peter Hall's Lecture: Steve Marron

Chair: Regina Liu
Room: Akamas A

Peter Hall and High Dimension Low Sample Size Asymptotics

10:00 - 11:00 Contributed Paper Session 1

Bayesian inference

Chair: Matteo Giordano
Room: Akamas C

10:00 **A Bayesian Nonparametric Approach To Super-Resolution Single-Molecule Localization**

Mariano Gabitto, Michael Jordan, Herve Marie-Nelly, Xavier Darzacq, Ari Pakman

10:20 **Nonparametric Bayesian inference for reversible multi-dimensional diffusions**

Matteo Giordano

Survival analysis I

Chair: Ali Shariati
Room: Aphrodite A

10:00 **Quantile Regression for Interval-Censored Data using Laguerre Polynomials**

Benjamin Dektelaere, Ingrid Van Keilegom, Anouar El Ghouch

10:20 **Challenges with model and tuning parameter selection when using cross-validation with censored data**

Anders Munch

10:40 **Asymptotic Behaviour of the MRL Function Estimator with Length-biased and Right Censored Data**

Ali Shariati, Hassan Doosti, Vahid Fakoor

Robust statistics I

Chair: Alexandre Lecestre
Room: Aphrodite B

10:00 **A Data-Driven Strategy for Specifying Cutoffs in Trimming and Winsorizing**

Derek Young, Kedai Cheng

10:20 **Robust estimation under a shape constraint**

Hélène Halconruy, Yannick Baraud, Guillaume Maillard

10:40 **Robust estimation in finite state space hidden Markov models**

Alexandre Lecestre

Clustered data

Chair: Erin Sprünken
Room: Christian Barnard

10:00 **Leave-cluster-out technique and variance estimation when regressors are many**

Stanislav Anatolyev

10:20 **A generalized framework for rank-based analysis of cluster data in the several sample case**

Erin Sprünken



Model selection

Chair: Ulrike Schneider

Room: Leda

10:00 **Model selection for robust learning of mutational signatures under overdispersion**
Marta Pelizzola, Ragnhild Laursen, Asger Hobolth

10:20 **The Probability of Improved Prediction: a new concept in statistical inference**
Stijn Jaspers, Olivier Thas

10:40 **On the geometry of uniqueness and model selection of LASSO, SLOPE and related estimators**
Ulrike Schneider, Patrick Tardivel

11:00 - 11:30 Coffee Break

11:30 - 12:30 Special Invited Talk: Richard Nickl

Chair: Ricardo Cao

Room: Akamas A

Bayesian non-linear inverse problems: progress and challenges

12:30 - 13:30 Lunch Break

13:30 - 15:30 Invited Paper Session 1

Structural inference in high dimensional models

Organiser: Eduard Belitser

Chair: Eduard Belitser

Room: Akamas A

13:30 **Dimension Estimation using Random Connection Models**
Paulo Serra, Michel Mandjes

14:00 **Characterizing the Type 1-Type 2 Error Trade-off for SLOPE**
Cynthia Rush

14:30 **Optimal Bayesian classification for high dimensional data**
Subhashis Ghoshal

15:00 **General framework for projection structures**
Eduard Belitser

Topics in curve estimation

Organiser: Anton Schick

Chair: Uschi Müller

Room: Akamas C

13:30 **Multivariate, Heteroscedastic Empirical Bayes via Nonparametric Maximum Likelihood**
Bodhisattva Sen, Jake Soloff, Adityanand Guntuboyina

14:00 **Learning high-dimensional functions by conditional least squares**
Stefano Vivogna

14:30 **Nonparametric inference for general categorical time series with time-varying parameters**
Lionel Truquet

15:00 **Location estimation, empirical Bayes, score matching, and their connection**
Min Xu



New directions in Bayesian nonparametric modeling

Organiser: Antonio Lijoi

Chair: Antonio Lijoi

Room: Aphrodite A

13:30 **Bayesian nonparametric methods for conditional independence testing**

Sarah Filippi

14:00 **Exact inference for a class of non-linear hidden Markov models on general state spaces**

Guillaume Kon Kam King, Matteo Ruggiero, Omiros Papaspiliopoulos

14:30 **Compound vectors of subordinators and their associated positive Lévy copulas**

Fabrizio Leisen, Alan Riva Palacio

15:00 **Nonparametric priors for partially exchangeable data: dependence structure and borrowing of information**

Beatrice Franzolini, Antonio Lijoi, Igor Pruenster, Giovanni Rebaudo

Recent advances in functional and complex data analysis

Organiser: Paromita Dubey & Hanlin Shang

Chair: Philip Reiss

Room: Aphrodite B

13:30 **Functional depth: Recent progress and perspectives**

Stanislav Nagy

14:00 **Random cohort effects and age groups dependency structure for mortality modelling and forecasting: Mixed-effects time-series model approach**

Bo Wang, Ka Kin Lam

14:30 **Limiting laws for optimal transport plans on finite spaces**

Yoav Zemel

15:00 **Continuous-time multivariate analysis**

Philip Reiss, Biplab Paul

Inference for Complex Problem Settings

Organiser: Regina Liu

Chair: Regina Liu

Room: Christian Barnard

13:30 **Query-augmented Active Metric Learning**

Annie Qu

14:00 **Additive regression with parametric help**

Hyerim Hong, Young Kyung Lee, Byeong Uk Park

14:30 **Repro Samples Method for Finite- and Large-Sample Inferences**

Minge Xie

15:00 **Sufficient Variable Selection via Expected Conditional Hilbert-Schmidt Independence Criterion**

Chenlu (Tracy) Ke

Non-parametric statistical methods for extremes

Organiser: Claudia Kluppelberg

Chair: Claudia Kluppelberg

Room: Leda

13:30 **Neural Networks for Extreme Quantile Regression and Risk Assessment** *Olivier Pasche, Sebastian Engelke*

14:00 **High-dimensional extreme quantile regression using partially-interpretable neural networks** *Jordan Richards, Raphael Huser*

14:30 **Changes in the distribution of observed annual maximum temperatures in Europe**

Ioannis Papastathopoulos, Graeme Auld, Gabriele Hegerl

15:00 **Max-linear Bayesian networks**

Claudia Klueppelberg

Recent advances in time series

Organiser: Giovanni Motta

Chair: Giovanni Motta

Room: Athena

13:30 **High-dimensional dynamic factor models: A selective survey and lines of future research.**
Manfred Deistler, Marco Lippi, Brian D.O. Anderson

14:00 **Semiparametric modeling of multiple quantiles**
Alessandra Luati, Leopoldo Catania

14:30 **Asymptotics for Spherical Functional Autoregressions**
Alessia Caponera, Domenico Marinucci

15:00 **Sequential on-line detection and classification in 3D Computer Vision**
Olympia Hadjiliadis

15:30 - 16:00 Coffee Break

16:00 - 18:00 Invited Paper Session 2

Recent Advances in Time Series Analysis

Organiser: Jens-Peter Kreiss

Chair: Jens-Peter Kreiss

Room: Akamas A

16:00 **Testing equality of spectral density operators for functional linear processes**
Anne Leucht, Efsthios Paparoditis, Theofanis Sapatinas, Daniel Rademacher

16:30 **Statistical methods for the restoration of historical music recordings**
Rainer Dahlhaus

17:00 **A Bootstrap-Assisted Self-Normalization Approach to Inference in Cointegrating Regressions**
Carsten Jentsch, Karsten Reichold

17:30 **Simultaneous Inference for Autocovariances based on Bootstrap Methods**
Alexander Braumann, Jens-Peter Kreiss, Marco Meyer

Nonparametrics in health related problems

Organiser: Yanyuan Ma

Chair: Yanyuan Ma

Room: Akamas C

16:00 **Multi-institutional Data for Real World Evidence**
Tianxi Cai

16:30 **Marginalized frailty-based illness-death model: application to the UK-biobank survival data**
Malka Gorfine

17:00 **Estimating Disease Onset from Change Points of Markers Measured with Error**
Tanya Garcia, Unkyung Lee, Raymond Carroll, Yuanjia Wang, Karen Marder

17:30 **Testing for Heterogeneity in the Utility of a Surrogate Marker**
Layla Parast, Tianxi Cai, Lu Tian

New insights in bandwidth selection

Organiser: Ricardo Cao

Chair: Ricardo Cao

Room: Aphrodite A

16:00 **Bandwidth selection for modal clustering**
Alessandro Casa, José E. Chacón, Giovanna Menardi



- 16:30 **Bandwidth selection for statistical matching and prediction**
Stefan Sperlich, Inés Barbeito, Ricardo Cao
- 17:00 **Bootstrap bandwidth selection for nonparametric default probability estimation: application to credit risk**
Rebeca Pelaez, Ricardo Cao, Juan Vilar
- 17:30 **Bagging cross-validated bandwidth selection in nonparametric regression estimation with application to Big Data**
Daniel Barreiro Ures, Ricardo Cao, Mario Francisco Fernández

Multivariate and Functional Time series

Organiser: Marco Meyer

Chair: Marco Meyer

Room: Aphrodite B

- 16:00 **Pivotal tests for relevant differences in the second order dynamics of functional time series**
Anne van Delft, Holger Dette
- 16:30 **Testing Linearity for Network Autoregressive Models**
Mirko Armillotta, Konstantinos Fokianos
- 17:00 **Spectral Inference for Functional Time Series in Hilbert Space**
Daniel Rademacher, Jens-Peter Kreiß, Efsthios Paparoditis
- 17:30 **A frequency domain bootstrap for general multivariate stationary processes**
Marco Meyer, Efsthios Paparoditis

Recent advance in covariance estimation and functional data analysis

Organiser: Jane-Ling Wang

Chair: Derek Young

Room: Christian Barnard

- 16:00 **Functional Data Analysis of Stochastic Differential Equations**
Victor Panaretos, Neda Mohammadi, Leonardo Santoro
- 16:30 **Joint non-parametric estimation of mean and auto-covariances for Gaussian processes**
Tatyana Krivobokova, Paulo Serra
- 17:00 **CoPE sets and Relevant Tubes**
Fabian Telschow, Junting Ren, Armin Schwartzman
- 17:30 **Interpoint-Ranking Sign Covariance for Test of Independence**
Kehui Chen

Statistical analysis of non-Euclidean and high-dimensional data

Organiser: Byeong Park

Chair: Byeong Park

Room: Leda

- 16:00 **Total Variation Regularized Frechet Regression for Metric-Space Valued Data**
Zhenhua Lin, Hans-Georg Mueller
- 16:30 **Antipodal Refection Depth (ARD) for Multivariate and Functional Data & Nonparametric Outlier Detection**
Regina Liu
- 17:00 **Nonparametric regression on Lie groups with measurement errors**
Jeong Min Jeon, Byeong Uk Park, Ingrid Van Keilegom
- 17:30 **Bounds for the asymptotic distribution of the likelihood ratio**
Andreas Anastasiou

19:00 - 20:30 Welcome Reception

9:00 - 11:00 Invited Paper Session3

Semiparametric models

Organiser: Ursula Müller & Valentin Patilea

Chair: Ursula Müller

Room: Akamas A

9:00 **Compositions of discrete random structures in Bayesian nonparametrics**, Filippo Ascolani,
Antonio Lijoi, Igor Pruenster, Giovanni Rebaudo

9:30 **Some recent results on semiparametric transformation models**
Natalie Neumeyer

10:00 **Wilks' Theorem for Models Defined by Conditional Moment Equations with Weakly Dependent Data**
Valentin Patilea

10:30 **Data integration in high dimension with multiple quantiles**
Ursula U. Müller

New trends in high dimensional robust statistics

Organiser: Mohamed Ndaoud

Chair: Mohamed Ndaoud

Room: Akamas C

9:00 **Coding convex bodies under Gaussian noise, and the Wills functional**
Jaouad Mourtada

9:30 **Robust leave-one-out cross-validation for high-dimensional Bayesian models**
Giacomo Zanella

10:00 **All-In-One Robust Estimator of the Gaussian Mean**
Arshak Minasyan

10:30 **Outlier detection in networks**
Olga Klopp, Geneviève Robin, Solenne Gaucher

Recent advances in change point and time series problems in high dimensions

Organiser: Hira Koul & Zhou Zhou

Chair: Liudas Giraitis

Room: Aphrodite A

9:00 **High-dimensional changepoint estimation with heterogeneous missingness**
Tengyao Wang, Bertille Follain, Richard Samworth

9:30 **Multiple change-points detection in generalized linear models**
Yuehua Wu

10:00 **Empirical process theory for stochastic processes using the functional dependence measure**
Stefan Richter

10:30 **Time-Varying Instrumental Variable Estimation**
Liudas Giraitis, George Kapetanios, Massimiliano Marcellino

Recent contributions in statistical analysis of nonstationary processes

Organiser: Anna Dudek

Chair: Anna Dudek

Room: Aphrodite B

9:00 **Some contributions to harmonizable time series analysis**
Jean-Marc Freyermuth



09:30 **Spectral Non-linear Granger Causality**

Hernando Ombao

10:00 **Spectral density estimation for nonstationary data with nonzero mean function**

Anna Dudek, Lukasz Lenart

Some recent advances in estimation and inference for complex data models

Organiser: Moulinath Banerjee

Chair: Moulinath Banerjee

Room: Christian Barnard

9:00 **Unlinked monotone regression**

Charles Doss, Fadoua Balabdaoui, Cécile Durot

9:30 **A General Modeling Framework for Network Autoregressive Processes**

George Michailidis

10:00 **Fast Network Community Detection with Profile-Pseudo Likelihood Methods**

Ji Zhu

Computer-intensive methods and dependent data

Organiser: Dimitris Politis

Chair: Dimitris Politis

Room: Leda

9:00 **Estimation and inference via the integrated copula spectrum**

Holger Dette, Stanislav Volgushev, Tobias Kley, Marc Hallin, Yuichi Goto, Ria Van Hecke

9:30 **Bootstrap for Dynamical Systems** Kasun Fernando, *Nan Zou*

10:00 **High-dimensional Change-point Detection Using Generalized Homogeneity Metrics**

Xianyang Zhang, Shubhadeep Chakraborty

10:30 **Rate of convergence in regenerative bootstrap**

Patrice Bertail, François Portier

Recent advances in semiparametric inference

Organiser: Christophe Ley

Chair: Olivier Thas

Room: Athena

9:00 **Experiences with the super learner**

Thomas Alexander Gerds

09:30 **Using Laguerre Polynomials in Semiparametric Estimation with an Application to Quantile Regression and Epidemics**

Alexander Kreiss, Ingrid Van Keilegom

10:00 **Semi-parametric quantile regression**

Anneleen Verhasselt, Ewnetu Worku, Irène Gijbel

10:30 **Semiparametric Methods for Covariate Adjustment for GPC Effect Sizes**

Olivier Thas

11:00 - 11:30 Coffee Break

11:30 - 12:30 Keynote Talk: Richard A. Davis

Chair: Dimitris Politis

Room: Akamas A

Statistical Learning of Multivariate Extremes

12:30 - 13:30 Lunch Break

13:30 - 15:30 Invited Paper Session 4

Recent advances in change point analysis and high-dimensional time series

Organiser: George Michailidis

Chair: George Michailidis

Room: Akamas A

13:30 **Estimation of High-Dimensional Markov-Switching VAR Models with an Approximate EM Algorithm**

Ali Shojaie, Xiudi Li, Abolfazl Safikhani

14:00 **Two-sample tests for relevant differences in the eigenfunctions of covariance operators**

Alexander Aue, Holger Dette, Gregory Rice

14:30 **Frequency-domain graphical models for multivariate time series**

Sumanta Basu

15:00 **Robust change point and change plane estimation in fixed and growing dimensions under heavy tailed errors**

Moulinath Banerjee

Recent innovations in Bayesian Nonparametrics

Organiser: Subhashis Ghoshal

Chair: Subhashis Ghoshal

Room: Akamas C

13:30 **Bayesian mode estimation and rate acceleration through a two-stage procedure**

William Weimin Yoo

14:00 **Characterization and Bayesian estimation of sparse regression parameters under linear inequality constraints**

Anindya Roy

14:30 **Bayesian sensitivity analysis for a missing outcomes model**

Stephanie van der Pas, Bart Eggen, Aad van der Vaart

Nonparametric approaches in Econometrics

Organiser: Enno Mammen

Chair: Enno Mammen

Room: Aphrodite A

13:30 **Flexible Covariate Adjustments in Regression Discontinuity Designs**

Claudia Noack, Christoph Rothe, Tomasz Olma

14:00 **Multiplicative Deconvolution In a Bivariate Stochastic Volatility Model**

Sergio Brenner Miguel

14:30 **Square-root-2 estimation for smooth eigenvectors of matrix-valued functions**

Giovanni Motta, Wei Biao Wu, Mohsen Pourahmadi

15:00 **Estimation of Group Structures in Individual Fixed Effects Models**

Enno Mammen, Ralf Wilke, Kristina Zapp

Resampling methods in non-standard situations

Organiser: Patrice Bertail

Chair: Patrice Bertail

Room: Aphrodite B

13:30 **Parameters on the boundary in predictive regression**

Iliyan Georgiev, Giuseppe Cavaliere

14:00 **General M-Estimator Processes and their m out of n Bootstrap with Functional Nuisance Parameters.**

Anouar Abdeldjaoued Ferfache

14:30 **Bootstrap on Null-recurrent Markov Chains**

Carlos Fernández

15:00 **Model-free Bootstrap and Conformal Prediction in Regression**

Dimitris Politis, Yiren Wang

Advances in functional data analysis

Organiser: Siegfried Hörmann

Chair: Siegfried Hörmann

Room: Christian Barnard

13:30 **Relative perturbation bounds with applications to empirical covariance operators**

Johannes Moritz Jirak

14:00 **Fast and Fair Simultaneous Confidence Bands for Functional Parameters**

Dominik Liebl

14:30 **Estimation of Functional ARMA Models**

Thomas Kuenzer

15:00 **On optimal prediction of missing functional data with memory**

Germain Van Bever, Lauri Viitasaari, Pauliina Ilmonen, Tommi Sottinen, Nourhan Shafik

New frontiers in network data analysis

Organiser: Marianna Pensky & Srijan Sengupta

Chair: Marianna Pensky

Room: Leda

13:30 **Partially-Exchangeable Multilayer Stochastic Block Models**

Daniele Durante, Francesco Gaffi, Antonio Lijoi, Igor Prünster

14:00 **Population-level Balance in Signed Networks**

Ji Zhu

14:30 **A nonparametric test of co-spectrality in networks**

Srijan Sengupta

15:00 **Discovering underlying dynamics in time series of networks**

Avanti Athreya, Zachary Lubberts, Youngser Park, Carey Priebe

Density and regression estimation under non standard conditions

Organiser: Valentin Patilea

Chair: Valentin Patilea

Room: Athena

13:30 **Empirical Risk Minimization under Random Censorship**

Stephan Clemençon

14:00 **Partly Linear Instrumental Variables Regression Without Smoothing on the Instruments**

Elia Lapenta, Jean-Pierre Florens

14:30 **Improved estimation of semiparametric cure regression models via presmoothing**

Eni Musta, Ingrid Van Keilegom, Valentin Patilea

15:00 **Semiparametric efficiency under identifiability constraints**

Mélanie Zetlaoui

15:30 - 16:00 Coffee Break



16:00 - 18:00 Invited Paper Session 5

New nonparametric and semiparametric methods on learning data with complex structures

Organiser: Wen Zhou

Chair: Chao Zheng

Room: Akamas A

16:00 **Confidence sets for Causal Discovery**

Mladen Kolar

16:30 **Data Integration Via Analysis of Subspaces (DIVAS)**

Jan Hannig

17:00 **Generalized dynamic factor models for spatio-temporal random fields on a network**

Chao Zheng

Expanding Statistical Frontiers with Nonparametric Methods

Organiser: Tanya Garcia

Chair: Tanya Garcia

Room: Akamas C

16:00 **Nonparametric instrumental regression with right censored duration outcomes**

Ingrid Van Keilegom, Jad Beyhum, Jean-Pierre Florens

16:30 **Robust and Efficient Estimation under Nonignorable Missing Response**

Yanyuan Ma

17:00 **Covariance and phase recovery for multivariate time series**

Irène Gannaz, Sophie Achard

17:30 **Bayesian variable selection based on empirical likelihood for ultra-high dimensional data**

Xinlei Wang, Can Xu, Yichen Cheng

New perspectives on nonparametric estimation of intensity and density functions

Organiser: María Dolores Martínez-Miranda

Chair: María Dolores Martínez-Miranda

Room: Aphrodite A

16:00 **Statistical learning for general point processes and applications to intensity and density estimation**

Christophe A.N. Biscio, Ottmar Cronie, Mehdi Moradi

16:30 **The modal age of Statistics**

José E. Chacón

17:00 **Kernel estimation methods beyond planar point processes**

María Isabel Borrajo García

17:30 **Superefficient estimation of future conditional hazards based on marker information**

Alex Isakson, Jens Nielsen, Enno Mammen, Cécile Proust-Lima

Financial Econometrics

Organiser: Genaro Sucarrat

Chair: Genaro Sucarrat

Room: Aphrodite B

16:00 **Inference on multiplicative component GARCH without any small-order moment**

Christian Francq, Baye Matar Kandji, Jean-Michel Zakoian

16:30 **Testing hypotheses on the innovations distribution in semi-parametric conditional volatility models**

Jean-Michel Zakoian, Christian Francq



17:00 **Subgeometrically ergodic autoregressions with autoregressive conditional heteroskedasticity**

Mika Meitz, Pentti Saikkonen

17:30 **Robust estimation and inference for time-varying unconditional volatility** Rickard Sandberg, Genaro Sucarrat

Nonparametric estimation

Organiser: Alexander Goldenshluger

Chair: Alexander Goldenshluger

Room: Christian Barnard

16:00 **Semi-parametric Bernstein-von Mises theorem for densities with jump under a mixture prior**

Natalia Bochkina, Judith Rousseau, J.-B. Salomond, Johan van der Molen Moris

16:30 **Statistical guarantees for high dimensional generative models**

Arnak Dalalyan, Victor-Emmanuel Brunel, Nicolas Schreuder

17:00 **Semiparametric estimation of McKean-Vlasov SDEs**

Denis Belomestny, Vytaute Pilipauskaite, Mark Podolskij

17:30 **Semiparametric ordered inference for conditional distributions**

Ori Davidov

Statistical methodology for complex and heterogeneous data

Organiser: Alexander Aue

Chair: Alexander Aue

Room: Leda

16:00 **Graphical models for nonstationary time series**

Suhasini Subbarao, Sumanta Basu

16:30 **Preprocessing functional data by a factor model approach**

Siegfried Hörmann, Fatima Jammoul

17:00 **Bootstrapping Spectral Statistics in High Dimensions**

Miles Lopes, Andrew Blandino, Alexander Aue

17:30 **Functional Sequential Treatment Allocation**

David Preinerstorfer, Anders Bredahl Kock, Bezirgen Veliyev

Exotic Testing and Inverse Problems

Organiser: Robert Mnatsakanov

Chair: Robert Mnatsakanov

Room: Athena

16:00 **What is the resolution of a microscope?**

Frank Werner

16:30 **Goodness of Fit Testing for Point Processes in Survival Analysis**

Umut Can, Roger Laeven, Estate Khmaladze

17:00 **Identification of convex polygons and non-negative probability measures on convex polygons from moment data**

Farhad Jafari, Robert Mnatsakanov, Satwik Pani

17:30 **A unified method of estimation in some statistical inverse problems**

Robert Mnatsakanov

18:15 - 19:15 General Meeting

Room: Akamas A

9:00 - 10:00 Keynote Talk: Aurore Delaigle

Chair: Marianna Pensky

Room: Akamas A

Estimation of the Distribution of Episodically Consumed Foods Measured with Error

10:00 - 11:00 Contributed Paper Session 2

Censored and missing data

Chair: Amichai Painsky

Room: Akamas A

10:00 **Nonparametric survival estimation with missing not at random censoring indicators**

Mikael Escobar-Bach, Olivier Goudet

10:20 **The mean, variance and correlation for bivariate recurrent event data with a terminal event**

Frank Eriksson

10:40 **Seeing the Unseen - a New Scheme for Missing Mass Estimation**

Amichai Painsky

High-dimensional data Chair: Nicolas Tavernier

Room: Akamas C

10:00 **Exponential bounds for regularized Hotelling statistics in high dimension**

El Mehdi Issouani, Patrice Bertail, Emmanuelle Gautherat

10:20 **Quantile LASSO with changepoints in panel data models**

Matus Maciak

10:40 **Nonlinear shrinkage estimation of large-dimensional covariance matrices using splines**

Nicolas Tavernier, Geert Dhaene

Copulas

Chair: Amitava Mukherjee

Room: Aphrodite A

10:00 **Tests of constant conditional dependence structures over partition sets**

Jean-David Fermanian, Alexis Derumigny, Aleksey Min

10:20 **Variable clustering using divergence type of dependence measures between random vectors**

Steven De Keyser, Irène Gijbels

10:40 **Nonparametric Surveillance of Unknown Structural Dependence in High-Dimensional Processes using Pseudo Copula and Eigenvectors – A Computational Approach**

Amitava Mukherjee

Screening and classification Chair: Dragan Radulovic

Room: Aphrodite B

10:00 **Novel nonparametric specification tests for additive concurrent model formulation**

Laura Freijeiro-González, Manuel Febrero-Bande, Wenceslao González-Manteiga

10:20 **Nonparametric Screening and Selection in presence of dependence among predictors with applications to environmental pollutants**

Sanjib Basu

10:40 **Accelerated Random Search for Weighted Nearest Neighbors**

Dragan Radulovic



Ranks and permutations Chair: Niels Olsen

Room: Leda

10:00 **A comparison of statistical methods for analyzing longitudinally measured ordinal outcomes in rare diseases**
Martin Geroldinger, Johan Verbeeck, Konstantin Emil Thiel, Geert Molenberghs, Arne Bathke, Georg Zimmermann

10:20 **Anomaly Detection for a Large Number of Streams: A Permutation/Rank-Based Higher Criticism Approach**
Ivo V. Stoepker, Rui M. Castro, Ery Arias-Castro, Edwin R. van den Heuvel

10:40 **Confidence regions for univariate and multivariate data using permutation tests**
Niels Olsen

Functional & Clustered Data

Chair: Tomas Masak

Room: Athena

10:00 **Convolutional Networks and Tensor Product ANOVA Smoothing Splines**
Zaid Harchaoui, Meyer Scetbon

10:20 **Nonparametric regression with clustered observations**
Stanislav Anatolyev

10:40 **Inference and Computation for Sparsely Sampled Random Surfaces**
Tomas Masak

Time series

Chair: Carina Beering

Room: Zeus

10:00 **Fully data-driven non-parametric estimation of Toeplitz covariance matrices**
Karolina Klockmann

10:20 **A Test of Independence for Locally Stationary Processes Using a Weighted Characteristic Function- based Distance**
Carina Beering

11:00 - 11:30 Coffee Break

11:30 - 12:30 Special Invited Talk: Victor Chernozhukov

Chair: Ingrid Van Keilegom

Room: Akamas A

Long Story Short: Omitted Variable Bias in Causal Machine Learning

12:30 - 13:30 Lunch Break

16:00 - 22:00 Excursion & Conference Dinner



9:00 - 11:00 Invited Paper Session 6

New issues in deconvolution problems

Organiser: Clément Marteau

Chair: Clément Marteau

Room: Akamas A

9:00 **Adaptive minimax testing for circular convolution***Sandra Schluttenhofer, Jan Johannes*9:30 **Deconvolution for some singular density errors with respect to the L1 loss***Mathieu Sart*10:00 **Rate optimal estimation of quadratic functionals in inverse problems with partially known operator***Martin Kroll*Statistics under nonstationarity

Organiser: Anne Leucht

Chair: Anne Leucht

Room: Akamas C

9:00 **Poisson Network Autoregression***Konstantinos Fokianos, Mirko Armillotta*9:30 **Bootstrap for integer-valued GARCH(p,q) processes***Michael Neumann*10:00 **Bootstrap confidence bands for the mean and covariance kernel of Banach space valued functional data***Melanie Birke, Christoph Reihl*10:30 **Learning to reflect***Lukas Trottnner*Topics in Econometrics: partial identification, boundary-adaptive kernel density estimation, and more TBA

Organiser: Jeff Racine

Chair: Jeff Racine

Room: Aphrodite A

09:00 **Estimating High Dimensional Monotone Index Models by Iterative Convex Optimization 1***Shakeeb Khan, Xiaoying Lan, Elie Tamer***On Optimal Set Estimation for Partially Identified Binary Choice Models***Shakeeb Khan, Tatiana Komorova, Denis Nekipelou*09:30 **Dynamic Ordered Panel Logit Models***Chris Muris, Bo Honoré, Martin Weidner*10:00 **Time-Varying Linear Transformation Models with Fixed Effects and Endogeneity for Short Panels***Irene Botosaru, Chris Muris, Senay Sokullu*10:30 **Boundary-Adaptive Kernel Density Estimation - Empirical Support Kernels and Cross-Validation: The Curious Case of the Uniform Density***Jeffrey Racine, Qi Li*



New developments in rank-based methods

Organiser: Michele La Rocca

Chair: Valeria Vitelli

Room: Aphrodite B

- 09:30 **Avoiding Degeneracies in Ordinal Unfolding Using Kemeny-Equivalent Dissimilarities for Two-Way Two-Mode Preference Rank Data**

Antonio D'Ambrosio

- 09:30 **Permutation testing for thick data when the number of variables is much greater than the sample size**
Patrick Langthaler, Riccardo Ceccato, Arne Bathke, Rosa Arboretti, *Luigi Salmaso*

- 10:00 **Ranks or Pseudo-Ranks? Some practical considerations**

Georg Zimmermann, Arne C. Bathke

- 10:30 **Pseudo-Mallows for Efficient Probabilistic Preference Learning**

Valeria Vitelli

New frontiers in network data analysis

Organiser: Marianna Pensky & Srijan Sengupta

Chair: Srijan Sengupta

Room: Christian Barnard

- 09:00 **Clustering of Diverse Multiplex Networks**

Marianna Pensky

- 9:30 **Avoiding a popularity contest: network clustering with the random walk Laplacian**

Alexander Modell

- 10:00 **Pseudo-likelihood-based M-estimation of networks with dependent connections and parameter vectors of increasing dimension**

Michael Schweinberger, Jonathan R. Stewart

Advances in semiparametric and nonparametric copula-based inference

Organiser: Irène Gijbels

Chair: Irène Gijbels

Room: Leda

- 9:00 **Bernstein estimators for conditional copulas and their derivatives**

Noel Veraverbeke

- 9:30 **A universal representation of statistical dependence**

Gery Geenens

- 10:00 **Kernel estimation of copulas for circular distributions**

Jose Ameijeiras-Alonso, Irène Gijbels

- 10:30 **Flexible multivariate distributions with central symmetry**

Jonas Baillien, Irène Gijbels, Anneleen Verhasselt

Nonparametric statistics and geometry

Organiser: Athanasios Georgiadis

Chair: Athanasios Georgiadis

Room: Athena

- 9:00 **Nonparametric estimation of covariance and autocovariance operators on the sphere**

Alessia Caponera, Julien Fageot, Matthieu Simeoni, Victor M. Panaretos

- 9:30 **Density estimation on manifolds or more general metric spaces**

Galatia Cleanthous, A..G. Georgiadis, G. Kerkycharian, P. Petrushev, D. Picard

- 10:00 **On high-frequency limits of U-statistics in Besov spaces over compact manifolds**

Claudio Durastanti, Domenico Marinucci, Xiaohong Lan

- 10:30 **On the rate of estimation for the invariant density of stochastic differential equations.**

Chiara Amorino, Arnaud Gloter



11:00 - 11:30 Coffee Break

11:30 - 12:30 Keynote Talk: Martin Wainwright

Chair: Alexander Goldenshluger

Room: Akamas A

Non-parametric estimation for reinforcement learning

12:30 - 13:30 Lunch Break

13:30 - 15:30 Invited Paper Session 7

Nonparametric Testing

Organiser: Miguel Delgado

Chair: Miguel Delgado

Room: Akamas A

13:30 **On the T-test**

Sergei. Novak

14:00 **Robust Inference On Infinite And Growing Dimensional Time Series Regression**

Abhimanyu Gupta, Myung Hwan Seo

14:30 **Debiased Semiparametric U-Statistics With an Application to Inequality of Opportunity**

Juan Carlos Escanciano

15:00 **Pearson Tests for Conditional Distributions.**

Miguel A Delgado, Julius Vainora

Modern challenges in structured time series data

Organiser: Yi Yu

Chair: Yi Yu

Room: Akamas C

13:30 **Adversarially robust change point detection**

Mengchu Li, Yi Yu

14:00 **Network Estimation by Mixing: Adaptivity and More**

Can Le, Tianxi Li

14:30 **Random Subspace Ensemble**

Yang Feng

15:00 **Optimal partition recovery: from chain graphs to lattices then general graphs.**

Yi Yu

Nonparametric random coefficient regression Organiser: Hajo Holzmann

Chair: Hajo Holzmann

Room: Aphrodite A

13:30 **Rate-optimal nonparametric estimation for random coefficient regression models**

Hajo Holzmann, *Alexander Meister*

14:00 **Random coefficients when regressors have limited variation** Christophe Gaillac, *Eric*

Gautier

14:30 **Tests for Qualitative Features in the Random Coefficients Model**

Katharina Proksch, Fabian Dunker, Konstantin Eckle, Johannes Schmidt-Hieber

15:00 **Varying Random Coefficient Models**

Christoph Breunig



New developments of semi- and nonparametric methods in biostatistics

Organiser: Ronghui Xu

Chair: Ronghui Xu

Room: Aphrodite B

- 13:30 **Statistical modelling of COVID-19 data: Putting generalised additive models to work**
Ursula Berger
- 14:00 **Doubly Robust Estimation under the Marginal Structural Cox Model for a Binary Treatment**
Denise Rava, Ronghui Xu, Jelena Bradic
- 14:30 **Semiparametric Estimation for Non-randomly Truncated Data**
Ronghui Xu, Yuyao Wang, Andrew Ying

OODA: Trees and Graph Structured Objects

Organiser: Steve Marron

Chair: Steve Marron

Room: Christian Barnatd

- 13:30 **Regression Model with Unlabelled Network Response on Graph Space.**
Anna Calissano, Aasa Feragen, Simone Vantini
- 14:00 **Manifold valued data analysis of samples of Networks**
Katie Severn, Ian Dryden, Simon Preston
- 14:30 **Null hypothesis testing for network-valued data with a multiscale approach: analysis of brain networks of patients with autism.**
Alessia Pini, Ilenia Lovato, Stamm Aymeric, Maxime Taquet, Simone Vantini
- 15:00 **Finite sample smeariness of Fréchet means on manifolds**
Stephan Huckemann, Benjamin Eltzner, Shayan Hundrieser

Data Analysis on Stratified Spaces with applications to SARSCov2 RNA analysis

Organiser: Vic Patrangenaru

Chair: Vic Patrangenaru

Room: Leda

- 13:30 **Wald Space for Statistics of Phylogenetic Trees**
Jonas Lueg, Stephan F. Huckemann, Tom M. W. Nye, Maryam K. Garba
- 14:00 **Investigating Two Possible Origins of SARS-CoV-2: an RNA Analysis on Tree Spaces**
Roland Moore, Victor Patrangenaru, Adam Dixon
- 14:30 **CLT on Stratified Spaces with an example of phylogenies of SARS-CoV-2 data**
Vic Patrangenaru, Chen Shen

Statistical Inference in Models with Functional Data

Organiser: Wenceslao Gonzalez Manteiga

Chair: Pedro Galeano

Room: Athena

- 13:30 **Bootstrap test for the equality of distributions in separable Hilbert spaces**
Ana Colubi, Gil Gonzalez-Rodriguez, Wenceslao Gonzalez-Manteiga, Manuel Febrero-Bande
- 14:00 **On the complexity index of a functional time series**
Kwo Lik (Lax) Chan, Aldo Goia, Enea Bongiorno
- 14:30 **A goodness-of-fit test for functional time series with applications to diffusion processes**
Javier Alvarez-Liéban, Wenceslao González-Manteiga, Alejandra López-Pérez, Manolo Febrero-Bande
- 15:00 **Goodness-of-fit tests for the functional linear model with scalar response with responses missing at random**
Manuel Febrero-Bande, Pedro Galeano, Eduardo García-Portugués, Wenceslao González-Manteiga

15:30 - 16:00 Coffee Break

16:00 - 18:00 Invited Paper Session 8

Nonparametric approaches in survival analysis

Organiser: Ingrid Van Keilegom

Chair: Ingrid Van Keilegom

Room: Akamas A

16:00 **Instrumental variable estimation of dynamic treatment effects on a survival outcome**

Jad Beyhum, Samuele Centorrino, Jean-Pierre Florens, Ingrid Van Keilegom

16:30 **A new lack-of-fit test for censored quantile regression models when the covariate is high-dimensional**

Mercedes Conde-Amboage, Ingrid Van Keilegom, Wenceslao González-Manteiga

17:00 **Causal inference for semi-competing risks data**

Daniel Nevo, Malka Gorfine

Bayesian Nonparametrics: theory and computation

Organiser: Sergios Agapiou

Chair: Sergios Agapiou

Room: Akamas C

16:00 **Complexity of coordinate-wise inference algorithms for crossed random effects**

Omiros Papaspiliopoulos, Giacomo Zanella, Swanardip Ghosh

16:30 **A Bayesian Nonparametrics Analysis of Finite Element and Graphical Representations of Gaussian Processes**

Daniel Sanz-Alonso, Ruiyi Yang

17:00 **A unified construction for series representations and finite approximations of completely random measures.**

Xenia Miscouridou

17:30 **Linear methods for non-linear inverse problems**

Botond Szabo, Aad van der Vaart, Geerten Koers

Recent advances in nonparametric and semiparametric statistics

Organiser: Lan Wang

Chair: Irène Gijbels

Room: Aphrodite A

16:00 **Clustering on the Sphere: State-of-the-art and a Poisson kernel-based method**

Marianthi Markatou

16:30 **"Flexible Semi-Parametric Gaussian Process Classification".**

Dipak Dey, Zhiyong Hu

17:00 **Gaussian copulas adjusted for non- and semi-parametric regression**

Irène Gijbels, Yue Zhao, Ingrid Van Keilegom

Statistical Topological Data Analysis and Geometric Inference

Organiser: Johannes Krebs

Chair: Daniel Rademacher

Room: Aphrodite B

16:00 **Goodness-of-fit tests for spatial tessellations based on the persistence diagram**

Christian Hirsch, Johannes Krebs, Claudia Redenbach

16:30 **Large deviation principle for geometric and topological functionals and associated point processes**

Takashi Owada, Christian Hirsch

17:00 **Statistical Query Complexity of Manifold Estimation**

Eddie Aamari

17:30 **Topological Inference via Bootstrap Resampling**

Benjamin Roycraft



Extremes and Machine Learning

Organiser: Anna Kiriliouk
Chair: Ioannis Papastathopoulos
Room: Christian Barnard

16:00 **Extreme value statistics in semi-supervised models**
John Einmahl

16:30 **Bayesian semi-parametric modeling of jointly heteroscedastic extremes**
Karla Vianey Palacios Ramirez

17:00 **Extreme conditional expectile estimation in heavy-tailed heteroscedastic regression models**
Gilles Stupfler, Stéphane Girard, Antoine Usseglio-Carleve

Analysis of high-dimensional complex data

Organiser: Eugen Pircalabelu
Chair: Eugen Pircalabelu
Room: Leda

16:00 **Sufficient Dimension Reduction for high-dimensional settings**
Andreas Artemiou, Eugen Pircalabelu

16:30 **Stationary Subspace Analysis - A Statistical Perspective**
Klaus Nordhausen, Lea Flumian, Markus Martilainen

17:00 **Inference for the explained Gini coefficient: a penalized bootstrap procedure**
Alexandre Jacquemain, Cédric Heuchenne, Eugen Pircalabelu

17:30 **The Completion Of Covariance Kernels**
Kartik Waghmare

Topics in nonparametric inference and estimation

Organiser: Hira L. Koul & Indeewara Perera
Chair: Indeewara Perera
Room: Athena

16:00 **Testing for Restricted Stochastic Dominance under Survey Nonresponse with Panel Data**
Rami Tabri, Matthew Elias

16:30 **Seasonal Cyclical Models**
Natalia Bailey, Karim Abadir, Walter Distaso, Liudas Giraitis

17:00 **Optimal bias correction of the log-periodogram estimator of the fractional parameter: A jackknife approach**
Kanchana Nadarajah, Gael M Martin, Donald S Poskitt

17:30 **Specification tests for GARCH processes with nuisance parameters on the boundary**
Indeewara Perera

18:15 - 19:15 Poster Session – Room: Foyer

- Bounding the error of discretized Langevin algorithms for non-strongly log-concave targets, Arnak Dalalyan, Lionel Riou-Durand, Avetik Karagulyan
- Bayesian Nonparametric Methods for Comparative Judgement Models and Vulnerability Estimation in Developing Countries
Rowland Seymour, David Sirl, Simon Preston, James Goulding, Madeleine Ellis
- Mean Shrinkage Estimation for Diagonal Multivariate Natural Exponential Families
Nikolas Siapoutis, Bharath Sriperumbudur, Donald Richards
- A Gaussian model for survival data subject to dependent censoring and confounding
Gilles Crommen, Jad Beyhum, Ingrid Van Keilegom
- Advances in Multi-Output Quantile Regression
Miroslav Siman
- Partially linear additive models on symmetric positive-definite matrices and Lie groups
Changwon Choi, Byeong Uk Park, Zhenhua Lin
- Reconstruction of discretely sampled functional data with missing values Siegfried Hörmann, Maximilian Ofner

9:00 - 10:00 Special Invited Talk: Gabor Lugosi

Chair: Omiros Papaspiliopoulos
Room: Akamas A

Problems in network archaeology: root finding and broadcasting

10:00 - 11:00 Contributed Paper Session 3

Nonparametric regression

Chair: Leonie Selk
Room: Akamas C

10:00 **Nonparametric tests to detect trends based on theory of records. An application to the analysis of climate change**
Ana C Cebrian, Jorge Castillo, Jesús Asín

10:20 **Panel nonparametric regression with tensorial long short-term memory recurrent neural networks**
Andrej Srakar

10:40 **Nonparametric Regression and Classification with Functional, Categorical, and Mixed Covariates**
Leonie Selk, Jan Gertheiss

Survival analysis II

Chair: Ali Shariati
Room: Aphrodite A

10:00 **Instrumental variable quantile regression under random right censoring**
Lorenzo Tedesco, Jad Beyhum, Ingrid Van Keilegom

10:20 **Single-index mixture cure models. An application to a study of cardiotoxicity in breast cancer patients.**
Beatriz Piñeiro Lamas, Ricardo Cao, Ana López Cheda

10:40 **A Goodness-of-fit Test with Length-biased Data**
Ali Shariati, Vahid Fakoor, Mahboobeh Akbari

Statistical inference in complex models

Chair: Yann Issartel
Room: Aphrodite B

10:00 **Geometric-Median-of-Means in Non-Positive Curvature Spaces**
Ho Yun, Byeong Uk Park

10:20 **Bootstrap inference in functional linear regression models with scalar response**
Hyemin Yeon, Danieal Nordman, Xiongtao Dai

10:40 **The Seriation and 1D-localization problems in latent space models.**
Yann Issartel

Directional data Chair: Diego Bolón

Room: Christian Barnard

10:00 **Nonparametric regression estimation for a functional-circular model**
Andrea Meilan-Vila, Rosa M. Crujeiras, Mario Francisco-Fernández

10:20 **Nonparametric multimodal regression for a circular response**
Maria Alonso-Pena, Rosa M. Crujeiras

10:40 **A hybrid method for estimating highest density regions of directional data**
Diego Bolón, Rosa M. Crujeiras, Alberto Rodríguez-Casal

Robust Statistics II

Chair: Markus Neuhaeuser

Room: Leda

10:00 **Computation of the halfspace depth regions – exact algorithm***Petra Laketa, Stanislav Nagy, Vit Fojtik, Pavlo Mozharovskyi*10:20 **Usefulness of the square-root allocation rule for many-to-one comparisons under non-normality***Markus Neuhaeuser*

11:00 - 11:30 Coffee Break

11:30 - 12:30 Contributed Paper Session 4

Permutation tests

Chair: Nick Koning

Room: Akamas C

11:30 **Time series comparisons: a permutation approach***Stefano Bonnini, Michela Borghesi*11:50 **More Efficient Exact Permutation Tests***Nick Koning, Jesse Hemerik*Change point analysis

Chair: Florian Pein

Room: Aphrodite A

11:30 **Semiparametric Detection of Changepoints in Location, Scale and Copula***Gaurav Agarwal, Idris Eckley, Paul Fearnhead*11:50 **NP-FOCuS: a Nonparametric Approach for Online Changepoint Detection***Gaetano Romano, Idris Eckley, Paul Fearnhead*12:10 **Cross-validation for change-point regression: pitfalls and solutions***Florian Pein*Robust statistics III

Chair: Ragnhild Laursen

Room: Aphrodite B

11:30 **Robust estimation of a regression function in exponential families***Juntong Chen*11:50 **Robust estimation of non-negative matrix factorization for mutational signatures using a flexible parametrization***Ragnhild Laursen, Asger Hobolth, Lasse Maretty Sørensen*Applications of nonparametric inference

Chair: Prajamitra Bhuyan

Room: Christian Barnard

11:30 **Adaptive wavelet estimation of a latent variable model***Andrej Srakar, Marilena Vecco*11:50 **Sales Forecasting with Weather Data using Fuzzy Natural Logic***Tomas Tichy*12:10 **Analysing the causal effect of London cycle superhighways on traffic congestion***Prajamitra Bhuyan, Emma McCoy, Haojie Li, Daniel Graham*



Extreme values

Chair: Dora Prata Gomes
Room: Leda

- 11:30 **Semi-parametric weighted Hill estimators**
Frederico Caeiro, Ayana Mateus
- 11:50 **Nonparametric Asymptotic Confidence Intervals for Extreme Quantiles**
Samuel Maistre, Laurent Gardes
- 12:10 **Nonparametric resampling methods in the estimation of parameters of rare events**
Dora Prata Gomes, Helena Penalva, Sandra Nunes, Manuela Neves

12:30 - 13:30 Lunch Break

13:30 - 15:30 Invited Paper Session 9

Inference for Dependent Data

Organiser: Kostas Fokianos
Chair: Kostas Fokianos
Room: Akamas A

- 13:30 **High-Dimensional Mixed Models with Varying Coefficients and Functional Random Effects** Michael Law,
Ya'acov Ritov
- 14:00 **Monotonic Alpha-divergence Minimisation for Variational Inference**
Kamelia Daudel, Randal Douc, Francois Roueff
- 14:30 **Hold-out estimates of prediction models for Markov processes**
Joseph Rynkiewicz
- 15:00 **Testing for changes in the tail-index of Long Memory Stochastic Volatility time series**
Davide Giraudo, Annika Betken, Rafal Kulik

Nonparametric Approaches to Complex Data Problems

Organiser: Somnath Datta
Chair: Somnath Datta
Room: Akamas C

- 13:30 **A Bayesian nonparametric approach to causal mediation with multiple mediators**
Michael Daniels
- 14:00 **Structurally Sparse Bayesian Neural Networks**
Taps Maiti, Sanket Jantree, Shrijita Bhattacharya
- 14:30 **Semiparametric Analysis of Clustered Interval-Censored Survival Data using Soft Bayesian Additive Regression Trees**
Debajyoti Sinha, Piyali Basak, Antonio Linero
- 15:00 **Regression Analysis of a Future State Entry Time Distribution Conditional on a Past State Occupation in a Progressive Multistate Model**
Somnath Datta

Computational Statistics

Organiser: Stefan Sperlich
Chair: Stefan Sperlich
Room: Aphrodite A

- 13:30 **Nonparametric inference for big-but-biased data**
Ricardo Cao, Laura Borraro
- 14:00 **Local inference for data giants**
Gilles Cattani, Michael Scholz, Stefan Sperlich
- 14:30 **Multivariate nonparametric change point detection with Random Forests and other classifiers**
Solt Kovács, Malte Lonschien, Peter Bühlmann
- 15:00 **Sparse Deep Learning**
Johannes Lederer



Multivariate Non-Parametric Tests

Organiser: Simos Meintanis

Chair: Charl Pretorius

Room: Aphrodite B

13:30 **On the IPCW approach for testing independence**

Marija Cuparić, *Bojana Milošević*

14:00 **Optimal nonparametric testing of Missing Completely At Random, and its connections to compatibility**

Tom Berrett

14:30 **Inference procedures for models with multivariate elliptically symmetric stable Paretian laws**

Charl Pretorius, Simos Meintanis, John Nolan, Zhou Zhou

Functional estimation in Monte Carlo methods

Organiser: Sylvain Le Corff

Chair: Luc Lehericy

Room: Christian Barnard

13:30 **Diffusion Schrodinger Bridge and Score-Based Generative Modeling**

Valentin De Bortoli, James Thornton, Arnaud Doucet, Jeremy Heng

14:00 **Fundamental limits for learning hidden Markov model parameters**

Kweku Abraham, Elisabeth Gassiat, Zacharie Naulet

14:30 **Deconvolution with general and unknown noise distribution**

Luc Lehericy, Sylvain Le Corff, Elisabeth Gassiat

Stochastic networks, graphs, and random matrices

Organiser: Anirban Dasgupta

Chair: Eugen Pircalabelu

Room: Leda

13:30 **Efficiency Lower Bounds for Distribution-Free Nonparametric Tests Based on Optimal Transport**

Nabarun Deb, Bhaswar Bhattacharya, Bodhisattva Sen

14:00 **Network Regression and Supervised Centrality Estimation**

Junhui Cai, Dan Yang, Wu Zhu, Haipeng Shen, Linda Zhao

14:30 **Community detection on probabilistic graphical models with group-based penalties**

Gerda Claeskens, Eugen Pircalabelu

Recent advances in dimension reduction and functional data analysis

Organiser: Bing Li

Chair: Efstathia Bura

Room: Athena

13:30 **Statistical Inference For Functional Linear Quantile Regression**

Peijun Sang

14:00 **Nonparametric graphical models for high-dimensional functional data**

Eftychia Solea, Holger Dette

14:30 **Functional Single-Index Models and Gaussian Stein's Identity**

Bharath Sriperumbudur, Krishna Balasubramanian, Hans-Georg Mueller

15:00 **Sufficient dimension reductions for mixed predictors**

Efstathia Bura, Liliana Forzani, Pamela Llop, Rodrigo Garcia Arancibia, Diego Tomass

LAST MINUTE CHANGES

MONDAY 20/06/2022

1) 10:00-11:00 Robust Statistics I

10:40 Usefulness of the square-root allocation rule for many-to-one comparisons under non-normality (*moved from Robust Statistics II*)
Markus Neuhaeuser

2) 13:30 – 15:30 New directions in Bayesian nonparametric modelling

Cancelled --14:30 Compound vectors of subordinators and their associated positive Lévy copulas
Fabrizio Leisen, Alan Riva Palacio

3) 13:30 – 15:30 Inference for Complex Problem Settings

Cancelled -- 15:00 Sufficient Variable Selection via Expected Conditional Hilbert-Schmidt Independence Criterion
Chenlu (Tracy) Ke

4) 16:00 – 18:00 Statistical analysis of non-Euclidean and high-dimensional data

Cancelled --16:30 - Antipodal Refection Depth (ARD) for Multivariate and Functional Data & Nonparametric Outlier Detection
Regina Liu

TUESDAY 21/06/2022

1) 09:00 – 11:00 Semiparametric models

Cancelled --9:30 Some recent results on semiparametric transformation models
Natalie Neumeier

2) 16:00 – 18:00 New nonparametric and semiparametric methods on learning data with complex structures

Cancelled -- 16:00 Confidence sets for Causal Discovery
Mladen Kolar

3) 13:30 – 15:30 Nonparametric approaches in Econometrics

Cancelled -- 15:00 Estimation of Group Structures in Individual Fixed Effects Models
Enno Mammen, Ralf Wilke, Kristina Zapp

THURSDAY 23/06/2022

1) 18:15 - 19:15 Poster Session – Room: Foyer

Additional Paper:
A model-based approach to density estimation in sup-norm
Guillaume Maillard

2) 16:00 - 18:00 Extremes and Machine Learning

Additional Paper:
17:30 Cross validation for rare events.
Anass Aghbalou

3) 16:00 - 18:00 Extremes and Machine Learning

Cancelled -- Extreme conditional expectile estimation in heavy-tailed heteroscedastic regression models
Gilles Stupfler, Stéphane Girard, Antoine Usseglio-Carleve

FRIDAY 24/06/2022

1) 10:00 – 11:00 Robust Statistics II

10:20 Robust estimation in finite state space hidden Markov models
Alexandre Lecestre (*moved from Robust Statistics I*)

2) 10:00 – 11:00 Survival analysis II

Cancelled -- 10:40 A Goodness-of-fit Test with Length-biased Data
Ali Shariati, Vahid Fakoor, Mahboobeh Akbari